Microstates and Macroscopic Thermodynamics of a Binary System

Statistical Physics – Laboratory Practice 1

Section 2 / Group E

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Abstract

This practice studies a system of independent, unbiased binary elements using the language of microstates, macrostates, multiplicity and Boltzmann entropy. A Python program enumerates small systems, calculates binomial multiplicities, creates macrostate-distribution plots and evaluates probabilities and entropy differences. We then use analytical approximations to examine the macroscopic limit, $N \sim 10^{23}$. The results illustrate why macrostates close to equal numbers of heads and tails dominate the probability distribution and how this statistical behaviour gives rise to the Second Law of Thermodynamics.

Repository: GitHub project repository

1 Introduction

The system consists of N independent, unbiased coins. Each coin has two possible outcomes, heads (H) or tails (T), and every complete configuration is assumed to be equally probable. A *microstate* specifies the outcome of every coin, while a *macrostate* is specified here by the total number of heads, k.

There are 2^N possible microstates. The multiplicity of the macrostate with k heads is the number of ways to choose which k coins show heads:

$$\Omega(N,k) = \binom{N}{k} = \frac{N!}{k!(N-k)!}. \tag{1}$$

Since all microstates are equiprobable, the probability of a macrostate is

$$P_k = \frac{\Omega(N,k)}{2^N}. \tag{2}$$

The Boltzmann entropy associated with a macrostate is

$$S(k) = k_B \ln \Omega(N,k), \tag{3}$$

where $k_B = 1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ is Boltzmann's constant.

2 Computational Method

The project is implemented in Python with separate modules for microstate enumeration, combinatorics, entropy and performance tests. The main program calls these functions, checks the calculated multiplicities, saves numerical results and generates vector PDF figures. For large binomial coefficients, logarithmic calculations are used to avoid integer-to-floating-point overflow:

$$\ln \Omega(N,k) = \ln \Gamma(N+1) - \ln \Gamma(k+1) - \ln \Gamma(N-k+1). \tag{4}$$

For distributions over all k, a recurrence relation is also used:

$$\Omega(N,0) = 1, \qquad \Omega(N,k+1) = \Omega(N,k)\frac{N-k}{k+1}. \tag{5}$$

The recurrence avoids recalculating factorials for every macrostate. Unit tests compare the computational results with exact binomial values for small systems.

3 Computational Analysis of Microstates

3.1 Enumeration for $N = 7$

For seven coins, the program enumerates all $2^7 = 128$ microstates and groups them according to the number of heads. The multiplicity distribution is: The multiplicities sum to 128, as required by the total number of

Table 1: Macrostate multiplicities for seven coins.

Heads k	Tails $7-k$	$\Omega(7,k)$
0	7	1
1	6	7
2	5	21
3	4	35
4	3	35
5	2	21
6	1	7
7	0	1

microstates. The complete list of the 128 configurations is included in Appendix A.

3.2 Maximum N for Full Enumeration

Full enumeration requires visiting all 2^N microstates, so its computational cost grows exponentially with N. The program benchmarks both the original enumeration method and the optimized method, applying the five-minute limit specified in the practice. It records the maximum completed N, execution times and peak memory use.

Table 2: Five-minute full-enumeration benchmark results.

Method	Maximum N	Time (s)	Peak memory (MiB)
Original	*to be filled*	*from CSV*	*from CSV*
Optimized	*to be filled*	*from CSV*	*from CSV*

The maximum is hardware-dependent. Increasing N by one doubles the number of microstates, which quickly makes exhaustive enumeration impractical in both time and memory.

3.3 Macrostate Distributions for $N = 20, 50, 80, 100$

For each requested system size, the program calculates $\Omega(N,k)$ for all $k = 0,\ldots,N$. It verifies that $\sum_{k=0}^{N}\Omega(N,k) = 2^N$ and saves the values in a CSV file. Figure 1 compares the four multiplicity distributions on a logarithmic vertical scale. The four individual plots are shown in Figure 2.

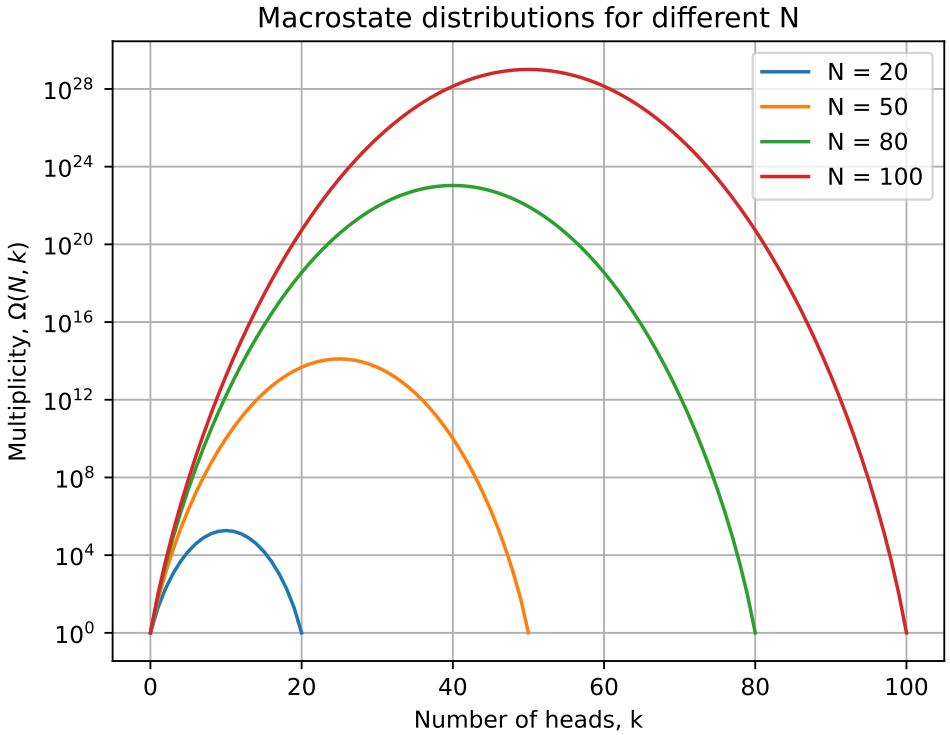
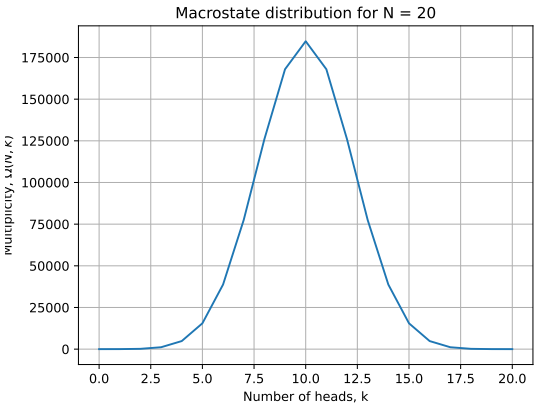
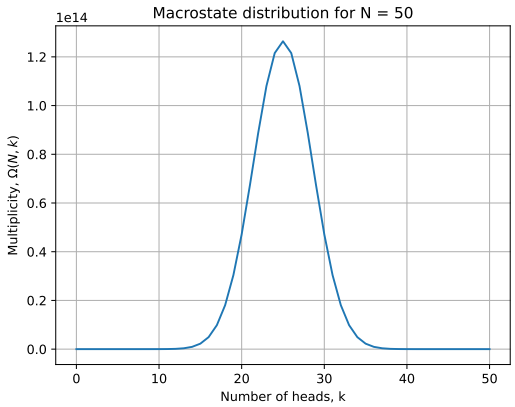


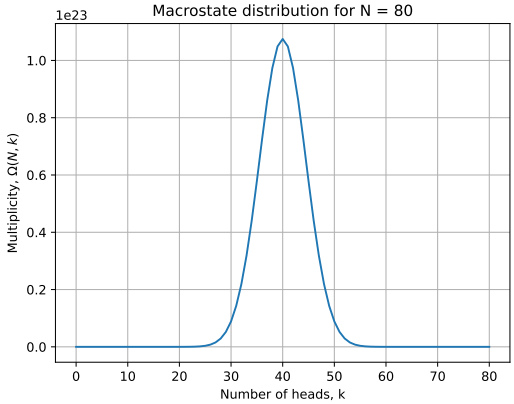
Figure 1: Macrostate multiplicities $\Omega(N, k)$ for $N = 20, 50, 80, 100$. The logarithmic vertical axis allows the four distributions to be compared on the same plot.



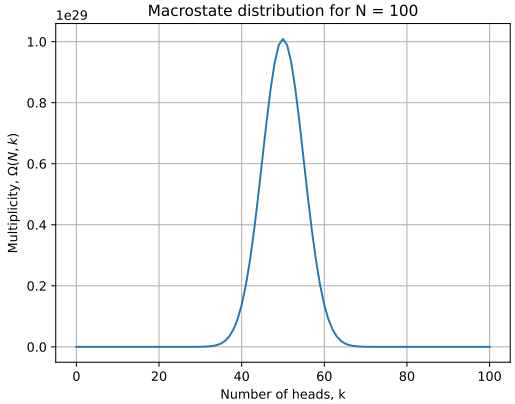
(a) $N = 20$



(b) $N = 50$



(c) $N = 80$



(d) $N = 100$

Figure 2: Individual macrostate multiplicities for the four system sizes. Each plot uses a linear vertical scale.

All distributions are symmetric about $k = N/2$, because $\Omega(N, k) = \Omega(N, N - k)$. Their maxima occur at the centre. As N increases, the peak multiplicity becomes larger and the distribution becomes wider in absolute number of heads.

3.4 Practical Limit for Distribution Calculation and Plotting

Unlike full microstate enumeration, calculating the complete multiplicity distribution requires only $N + 1$ macrostate values. The program tests the candidate system sizes

$$N = 100, 500, 1000, 2000, 5000, 10000, 20000, 50000, 100000,$$

measuring the calculation and plotting times and saving them in `results/distribution_scaling.csv`. The final tested value and its plot are recorded in `results/distribution_scaling_summary.txt`. Enter the largest value actually completed on the team's computer after turning on the distribution-test option in `main.py`.

Table 3: Practical distribution-calculation limit measured on the team's computer.

Largest tested N	Calculation time (s)	Plotting time (s)	Total time (s)
Pending	—	—	—

The limiting factor is the time and memory required to store and plot the $N + 1$ multiplicities, rather than the exponential enumeration of all 2^N microstates. The reported practical limit depends on the computer and the chosen test values.

3.5 Extrapolation to the Thermodynamic Limit

For unbiased coins, the number of heads follows a binomial distribution with mean and standard deviation

$$\langle k \rangle = \frac{N}{2}, \qquad \sigma_k = \frac{\sqrt{N}}{2}. \tag{6}$$

Near the centre, the binomial distribution is well approximated by a Gaussian:

$$P_k \simeq \sqrt{\frac{2}{\pi N}} \exp\left[-\frac{2(k - N/2)^2}{N}\right]. \tag{7}$$

The relative fluctuation is

$$\frac{\sigma_k}{\langle k \rangle} = \frac{1}{\sqrt{N}} \longrightarrow 0 \qquad (N \to \infty). \tag{8}$$

For $N = 10^{23}$, the mean is 5.0×10^{22} heads, the standard deviation is 1.5811×10^{11} heads, and the relative fluctuation is 3.1623×10^{-12}. Thus, the absolute width grows as $\sqrt{N}$, but its size relative to the system becomes negligible. This is the thermodynamic-limit concentration around equal numbers of heads and tails.

4 Probabilities, Entropy, and Thermodynamics

4.1 Team Parameter and Rounding Convention

The harmonic average specified in the practice gives

$$M = 31.2066, \qquad N = 100 + M = 131.2066. \tag{9}$$

Since the number of coins must be an integer, we round 131.2066 to the nearest integer and use $\boxed{N = 131}$ throughout the team-specific calculations. This rounding convention is stated explicitly.

For $N = 131$, which is odd, the two most probable macrostates are $k = 65$ and $k = 66$. They have equal multiplicities. We use $k = 66$ whenever a single central macrostate is needed. For the cumulative intervals, we include all integer values within the stated distance of the mean $N/2 = 65.5$.

Table 4: Probability calculations for the team system, $N = 131$.

Part	Macrostate or interval	Probability
(a)	Exactly $k = 30$ heads	1.2447×10^{-10}
(b)	Central macrostate, $k = 66$ (equal to $k = 65$)	6.9316×10^{-2}
(c)	$56 \le k \le 75$	9.1983×10^{-1}
(d)	$46 \le k \le 85$	9.9957×10^{-1}

4.2 Probability Calculations

Using Eq. (2), the probability results are shown in Table 4. Parts (c) and (d) are cumulative probabilities, obtained by summing all integer macrostates inside each interval.

4.3 Entropy Changes and Waiting Time

The entropy differences are defined as $\Delta S_{k_1,k_2} = S(k_1) - S(k_2)$. For parts (e) and (f), the target values $N/2 + 25 = 90.5$ and $N/2 + 10 = 75.5$ are rounded upward to $k = 91$ and $k = 76$, respectively. In part (i), the central macrostate $k = 66$ is compared with $k = 76$; the symmetric lower-side macrostate $k = 55$ has the same multiplicity as $k = 76$.

Table 5: Entropy and thermodynamic results for the team system, $N = 131$.

Part	Quantity	Result
(e)	$S(91) - S(66)$	-1.3957×10^{-22} J K^{-1}
(f)	$S(76) - S(66)$	-2.3109×10^{-23} J K^{-1}
(g)	$S(66) - S(20)$	$+4.7623 \times 10^{-22}$ J K^{-1}
(h)	$P(k = 8)$	6.3543×10^{-28}
(h)	Mean waiting time, $1/P(k = 8)$	1.5737×10^{27} s
(i)	$\Omega(131, 66)/\Omega(131, 76)$	5.3321

For part (h), a tossing rate of N coins per second means that one complete configuration of all N coins is generated each second. The mean number of independent trials required to obtain exactly eight heads is $1/P(k = 8)$, so the expected waiting time is the same number in seconds.

4.4 Large-N Limit: $N \sim 10^{23}$

We now reconsider parts (a)–(j) for $N = 10^{23}$. Direct enumeration is impossible because there are 2^N microstates. We instead use analytical approximations. For this even value of N, the central macrostate is exactly $k = N/2$.

For a fixed, small number k of heads,

$$\binom{N}{k} = \frac{N(N-1)\cdots(N-k+1)}{k!} \sim \frac{N^k}{k!}, \qquad P_k \sim \frac{N^k}{k!\,2^N}. \tag{10}$$

Consequently, it is more useful to report the logarithm of the probability for exactly 30 heads:

$$\log_{10} P(k = 30) \simeq -3.0103 \times 10^{22}. \tag{11}$$

For the central macrostate, Stirling's approximation gives

$$P(k = N/2) \simeq \sqrt{\frac{2}{\pi N}} = 2.5231 \times 10^{-12}. \tag{12}$$

For a fixed integer $m \ll \sqrt{N}$, the $2m + 1$ probabilities in the interval $N/2 - m \le k \le N/2 + m$ are practically equal to the central probability. Thus,

$$P\left(\left|k - \frac{N}{2}\right| \le m\right) \simeq (2m + 1)\sqrt{\frac{2}{\pi N}}. \tag{13}$$

For an offset m from the centre, the entropy difference is approximately

$$S\left(\frac{N}{2}+m\right)-S\left(\frac{N}{2}\right)\simeq-\frac{2k_Bm^2}{N}. \tag{14}$$

For part (g), comparison with the macrostate containing only 20 heads gives, to leading order,

$$S\left(\frac{N}{2}\right)-S(20)\simeq Nk_B\ln 2. \tag{15}$$

Table 6: Analytical results for the macroscopic system, $N=10^{23}$. Logarithms are reported for quantities outside a useful numerical range.

Part	Quantity	Result
(a)	$\log_{10}P(k=30)$	-3.0103×10^{22}
(b)	$P(k=N/2)$	2.5231×10^{-12}
(c)	$P(\lvert k-N/2\rvert\le10)$	5.2986×10^{-11}
(d)	$P(\lvert k-N/2\rvert\le20)$	1.0345×10^{-10}
(e)	$S(N/2+25)-S(N/2)$	-1.7258×10^{-43} J K^{-1}
(f)	$S(N/2+10)-S(N/2)$	-2.7613×10^{-44} J K^{-1}
(g)	$S(N/2)-S(20)$	$+9.5699\times10^{-1}$ J K^{-1}
(h)	$\log_{10}(t_{\mathrm{wait}}/\mathrm{s})$ for exactly 8 heads	3.0103×10^{22}
(i)	$\Omega(N,N/2)/\Omega(N,N/2+10)$	$1+2.0\times10^{-21}$

For part (h), we again assume one complete N-coin trial per second. The expected waiting time is so large that its base-10 logarithm is the most informative representation. For part (i), the likelihood ratio approaches one because a fixed displacement of ten heads is negligible compared with the distribution width when N is macroscopic. More precisely, its logarithm is approximately $2(10)^2/N$.

The natural width of the distribution is

$$\sigma_k=\frac{\sqrt{N}}{2}=1.5811\times10^{11}\text{ heads},\qquad\frac{\sigma_k}{N/2}=\frac{1}{\sqrt{N}}=3.1623\times10^{-12}. \tag{16}$$

Therefore, windows of only 10 or 20 heads are extremely narrow compared with the typical fluctuation scale. The central macrostate is the most probable individual macrostate, but its probability tends to zero as $N^{-1/2}$. Most of the total probability is spread over a number of macrostates of order $\sqrt{N}$.

4.5 Second Law Interpretation

None of the results contradicts the Second Law of Thermodynamics. In this model, the macrostate entropy is determined by the number of microstates compatible with that macrostate. A macrostate near $N/2$ has a much larger multiplicity than a strongly ordered macrostate such as $k=20$, and is therefore much more probable. For finite N, entropy-decreasing fluctuations are not forbidden: they have a non-zero probability. The Second Law is statistical rather than an absolute prohibition of such fluctuations. As N becomes macroscopic, the probability of strongly ordered macrostates becomes extraordinarily small, while relative fluctuations vanish as $1/\sqrt{N}$. This explains why macroscopic systems overwhelmingly evolve toward behaviour associated with higher entropy.

5 Conclusion

The coin model shows how a simple set of independent binary elements produces a strongly peaked macrostate distribution. Exact enumeration is practical only for small N, whereas multiplicities can be calculated for much larger systems without listing every microstate. For $N\sim10^{23}$, the distribution is concentrated near equal numbers of heads and tails in relative terms, despite having a large absolute fluctuation width. The results give a statistical interpretation of entropy and explain why the Second Law is overwhelmingly reliable for macroscopic systems.

A Complete Microstate List for $N = 7$

The following list contains every seven-coin configuration, grouped by the number of heads. It is the complete set of $2^7 = 128$ microstates generated by the program.

0 heads, 7 tails (1 microstate):
TTTTTTT

1 head, 6 tails (7 microstates):
HTTTTTT, THTTTTT, TTHTTTT, TTTHTTT, TTTTHTT, TTTTTHT, TTTTTTH

2 heads, 5 tails (21 microstates):
HHTTTTT, HTHTTTT, HTTHTTT, HTTTHTT, HTTTTHT, HTTTTTH, THHTTTT, THTHTTT, THTTHTT, THTTTHT, THTTTTH, TTHHTTT, TTHTHTT, TTHTTHT, TTHTTTH, TTTHHTT, TTTHTHT, TTTHTTH, TTTTHHT, TTTTHTH, TTTTTHH

3 heads, 4 tails (35 microstates):
HHHTTTT, HHTHTTT, HHTTHTT, HHTTTHT, HHTTTTH, HTHHTTT, HTHTHTT, HTHTTHT, HTHTTTH, HTTHHTT, HTTHTHT, HTTHTTH, HTTTHHT, HTTTHTH, HTTTTHH, THHHTTT, THHTHTT, THHTTHT, THHTTTH, THTHHTT, THTHTHT, THTHTTH, THTTHHT, THTTHTH, THTTTHH, TTHHHTT, TTHHTHT, TTHHTTH, TTHTHHT, TTHTHTH, TTHTTHH, TTTHHHT, TTTHHTH, TTTHTHH, TTTTHHH

4 heads, 3 tails (35 microstates):
HHHHTTT, HHHTHTT, HHHTTHT, HHHTTTH, HHTHHTT, HHTHTHT, HHTHTTH, HHTTHHT, HHTTHTH, HHTTTHH, HTHHHTT, HTHHTHT, HTHHTTH, HTHTHHT, HTHTHTH, HTHTTHH, HTTHHHT, HTTHHTH, HTTHTHH, HTTTHHH, THHHHTT, THHHTHT, THHHTTH, THHTHHT, THHTHTH, THHTTHH, THTHHHT, THTHHTH, THTHTHH, THTTHHH, TTHHHHT, TTHHHTH, TTHHTHH, TTHTHHH, TTTHHHH

5 heads, 2 tails (21 microstates):
HHHHHTT, HHHHTHT, HHHHTTH, HHHTHHT, HHHTHTH, HHHTTHH, HHTHHHT, HHTHHTH, HHTHTHH, HHTTHHH, HTHHHHT, HTHHHTH, HTHHTHH, HTHTHHH, HTTHHHH, THHHHHT, THHHHTH, THHHTHH, THHTHHH, THTHHHH, TTHHHHH

6 heads, 1 tail (7 microstates):
HHHHHHT, HHHHHTH, HHHHTHH, HHHTHHH, HHTHHHH, HTHHHHH, THHHHHH

7 heads, 0 tails (1 microstate):
HHHHHHH

B AI Collaboration and Verification Statement

AI tools were used as coding and writing assistants during the project. The team is responsible for reviewing, understanding and verifying every submitted calculation and explanation. The Python implementation is tested with `pytest`; the team should retain its prompt log and verification notes in `AI_STATEMENT.md`, as required by the project instructions.

C Reference

V. Tribaldos, *Microstates of a System of Coins: First Laboratory Practice*, Statistical Physics, Universidad Carlos III de Madrid, Academic Year 2026–2027.